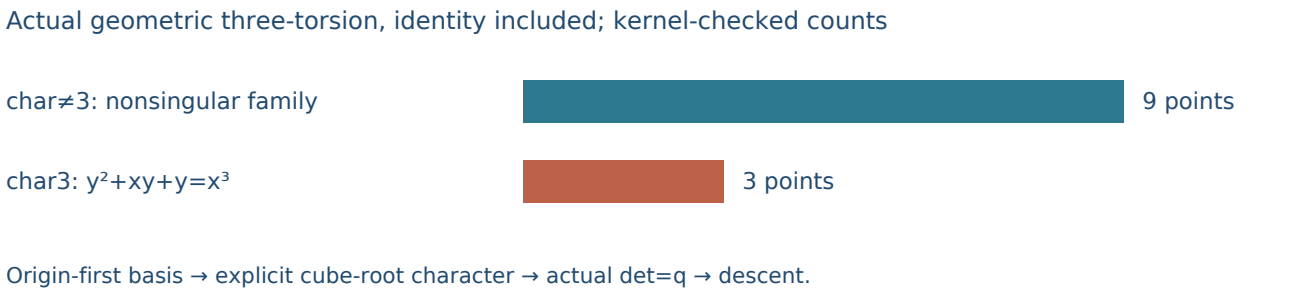
A cubic character proves actual Frobenius det=q

Result. Lean proves the actual Frobenius determinant formula at prime3 for every nonsingular model $y^2+a1xy+a3y=x^3$ over every finite field of characteristic other than three, including two. It derives the qualified BK containment criterion for every chosen finite extension and compatible embedding, with $q\neq1 \bmod 3$. No determinant, rank, pairing or abstract rationality premise is supplied. Arbitrary-curve normalization remains separate. Signed standing unverified. [pcc:unverified]



Actual character. P=(0,0) is a nonzero actual point killed by three; $a3\neq0$ follows from the discriminant. For an actual three-torsion point Q=(x,y) outside the origin line, define $w(Q)=-(2y+a1x+a3)/(y+a1x+2a3)$. The literal division polynomial, curve equation and nonsingularity prove the denominator is nonzero and $w^2+w+1=0$. Actual addition of P preserves the value and actual negation inverts it. Set the value to1 on O, P and−P; that is exactly its kernel. These statements hold without division by two. The normal form is not assumed to have exactly a cyclic full torsion group. [pcc:unverified]

Actual determinant and extension images. Reuse the preceding general geometric rank/basis proof, then construct an actual basis with P first by an invertible F3 matrix and actual linear equivalence. The character is the second basis vector’s primitive cube root raised to the actual second coordinate. Actual point Frobenius fixes P and sends character values to their q-th powers. Primitive-root exponent injectivity forces the second diagonal entry to be qmod3; the first column is(1,0), so the determinant isq. Basis independence gives every basis. Existing actual chosen-field descent and the qualified matrix theorem then prove the source criterion in this family. [pcc:unverified]

Controls. The genuine nonsingular characteristic3 model $y^2+xy+y=x^3$ has discriminant1, exactly three geometric three-torsion points and actual module rank1, using an explicit subgroup/OptionBool equivalence. All three already lie in every compatible extension image, while3never divides3^k−1 for positivek. Lean directly refutes the unrestricted inclusion paraphrase after deleting coprimality. Original BK asks for nine points in this case; that condition is false too, so its original theorem is not refuted. Separate singular controls distinguish an equation point from a nonsingular group point and show nonzeroa3does not alone force nonzero discriminant. [pcc:unverified]

Sources and remaining scope. BK1998 Theorem1/proof/Remarks pp142–143, DOI10.1007/s001459900040; Dujella author-hosted section15.3,p494 coordinate/order3normal form; Elkies HarvardMath2232024 III.8 tangent functions and pairing-convention warning; Snowden Michigan2013Lecture2 Weil/determinant passage. Selected sections read, exact primary hashes and inspected p494 retained. Prior general rank/Frobenius/descent and F5/F7/F25 cases credited. General coordinate/group/Frobenius normalization, canonical Weil identification, other primes and full all-k p<500census remain open. No full-book/paper read, novelty, qualified-source refutation or corpus promotion. [pcc:unverified]

Validation. Whole gate77/0failures/0trusted; complete statement binding and recursive custom-source/config pins. All209build-provenance files roundtrip through62unique objects; archive64558B and five gate files retained. Scalar IDC1h3cd3ab88de4dfcba, four variables/degree4: universal Lean identity and official32+32-point check, same producer/host; it does not certify geometry. Pinned Lean4.35.0/Mathlib56a00585…, only propext/Classical.choice/Quot.sound. No holes/custom axioms/native evaluator/raised limits/new global flags/waivers. Exact PCC replay and signed standing are separate. KB/Conductor unavailable, primary/repository/library fallback. No native experiment, paid compute or agent upload. [pcc:unverified]